

Cybersecurity

Historical ciphers

The Decrypted Letter: T H E D E C R Y P T E D L E T T E R

The Encrypted Letter: K L M N O P Q R S T U V W X Y Z A B C D E F G H I J

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Agenda

- Formal aspects
- Cipher systems
- Historical ciphers
- Transposition ciphers
- Substitution ciphers
- Product ciphers

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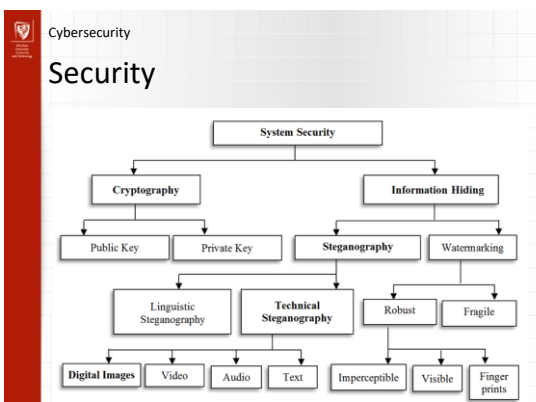
Crypto...

- Cryptography - ?
- Cryptanalysis - ?
- Cryptology - ?
- Steganography - ?

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Crypto...

- Cryptography
 - making secret or/and protecting messages – algorithms design
- Cryptanalysis
 - breaking secret messages/breaking algorithms
- Cryptology
 - cryptography and cryptanalysis
- Steganography
 - concealing messages



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Steganography

Hiding the fact of communication itself

- Concealing messages within other data

Classical methods:

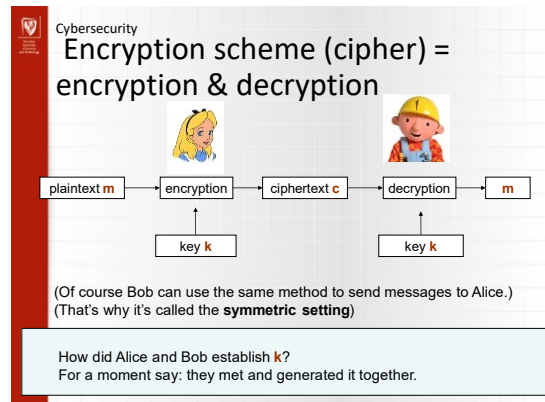
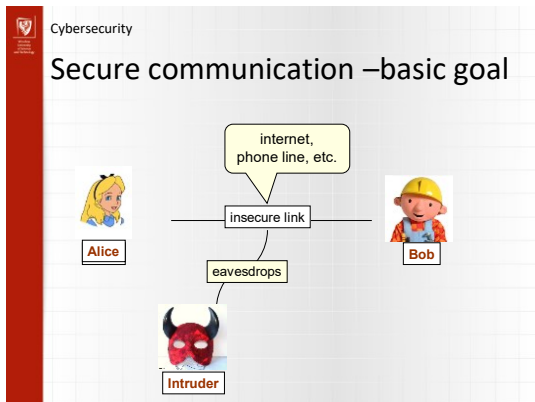
- Invisible ink
- Masking (grilles)
- Microscopic tattoos

Digital steganography:

- Hiding in least significant bits of pixels
- Modifying file headers

Herodotus (484-425 B.C.)

Persian War (~480 BC), Demaratus' secret message to the Spartans was hidden under the wax on a wooden tablet (steganography).



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Kerckhoffs' principle

Auguste Kerckhoffs (1883):
The enemy knows the system

- The cipher should remain secure even if **the adversary knows the specification of the cipher.**
- The only thing that is **secret** is a
 - short key k
 - that is **usually chosen uniformly at random**

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Kerckhoffs' principle – the motivation

- In commercial products:
 - it is unrealistic to assume that the design details remain secret (**reverse-engineering!**)

For example:

- In **1987** **RSA** designed a cipher called **RC4**.
- Its specification was a trade secret.
- In **1994** a C code of **RC4** was posted on a cypherpunk mailing list

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Kerckhoffs' principle

- The design details can be discussed and **analyzed in public.**
- It corresponds nicely to the **open-source** ideology.
- The schemes can become parts of **industrial standards**

Not respecting this principle
=
"security by obscurity".

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Cipher system

- Encryption domains and codomains
 - A denotes a finite set called **alphabet of definition**.
 - Example: $A = \{0,1\}$
 - M denotes a set called **message space**.
 - M consists of strings of symbols from an alphabet of definition
 - An element of M is called a **plaintext message** (or a **plaintext**)
 - Example: M may consist of binary strings, English text, etc.
 - C denotes a set called **ciphertext space**.
 - C consists of strings of symbols from an alphabet definition
 - C may differ from the alphabet of definition for M
 - An element of C is called a **ciphertext**

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Cipher system

- Encryption and decryption transformations
 - K denotes a set called the **key space**, an element of K is called a **key**
 - Each $e \in K$ uniquely determines a bijection from M to C , denoted by E_e (called **encryption function** or **encryption transformation**)
 - E_e must be a bijection if the process to be reversed and a unique plaintext recovered from each distinct ciphertext
 - Each $d \in K$, D_d (called **decryption function** or **decryption transformation**) denotes a bijection from C to M

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Cipher system

- The process of applying the transformation E_e to a plaintext $m \in M$, is referred to as the **encryption of m**
- The process of applying the transformation D_d to a ciphertext $c \in C$, is referred to as the **decryption of c**

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Cipher system

- An **encryption scheme** consists of (in addition to M, C, K):
 - A set $\{E_e : e \in K\}$ of encryption transformations
 - A correspondent set $\{D_d : d \in K\}$ of decryption transformations with the property that for each $e \in K$ there is a unique key $d \in K$ such that

$$D_d = E_e^{-1} \Leftrightarrow D_d(E_e(m)) = m$$
 - An encryption scheme is also referred to as a **cipher**
 - e and d are referred to as a **key pair** denoted by (e, d)
 - Encryption achieves Confidentiality

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Taxonomy of ciphers

- **Symmetric Key**
 - Same key for encryption and decryption or having one key is easy to find the other one
 - Two types: Stream ciphers, Block ciphers
- **Public Key** (or asymmetric crypto)
 - Two keys, one for encryption (public), and one for decryption (private)
 - It is difficult (computationally impossible) to find private key knowing the public key
 - digital signatures

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Historical Ciphers

- Historical (classical) cryptography:
 - Two basic types
 - Transposition ciphers
 - Substitution ciphers
 - Product ciphers
 - Combinations of the two basic types

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Historical Ciphers

Types of Cipher

The diagram illustrates two types of classical ciphers. The first is a Transposition Cipher, where the letters of the plaintext 'hello' are rearranged according to the key '23541' to produce the ciphertext 'ELOH'. The second is a Substitution Cipher, where each letter of the plaintext 'hello' is replaced by a corresponding letter in the ciphertext 'KHOOR'. Below the diagrams, the plaintext and cipher alphabets are listed: Plaintext alphabet: a b c d e f g h i j k l m n o p q r s t u v w x y z; Cipher alphabet: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C.



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Historical Ciphers - example

- Trivial substitution – shift cipher: (M, C, K, E, D)
 - $M = \{\text{sequences of letters}\}$
 - $C = M$
 - $K = \{i \mid i \text{ is an integer and } 0 \leq i \leq 25\}$
 - $E = \{E_k \mid k \in K \text{ and for all letters } m,$
 $E_k(m) = (m + k) \bmod 26\}$
 - $D = \{D_k \mid k \in K \text{ and for all letters } c,$
 $D_k(c) = (26 + c - k) \bmod 26\}$



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Historical Ciphers - example

- Let $k = 9$, $m = \text{"LECTURE"} (11, 4, 2, 19, 20, 17, 4)$
 - $E_k(m) = (m + k) \bmod 26$
 $= (20, 13, 11, 2, 3, 0, 13) \bmod 26$
 $= \text{"20,13,11,28,29,26,13"} = \text{"UNLCDAN"}$
 - $D_k(c) = (26 + c - k) \bmod 26$
 $= (37, 30, 28, 19, 20, 17, 30) \bmod 26$
 $= \text{"11,4,2,19,20,17,4"} = \text{"LECTURE"}$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25



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Security of a substitution cipher

- What is the size of key space in the substitution cipher assuming 26 letters?

$$|\mathcal{K}| = 26$$

$$|\mathcal{K}| = 26! \quad (26 \text{ factorial}) \quad 403291461126605635584000000$$

$$|\mathcal{K}| = 2^{26}$$

$$|\mathcal{K}| = 26^2$$



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Transposition Ciphers

- A **simple transposition cipher** with fixed period t :
 - Consists of grouping the plaintext into blocks of t characters, and applying to each block a single permutation e on the numbers 1 through t .
 - Decryption consists of the permutation d which inverts e
- **Compound transposition**
 - is a sequential composition of 2 or more simple transpositions with respective periods $t_1, \dots, t_r$



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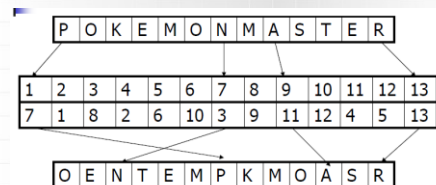
Transposition Ciphers

- A **simple transposition cipher** with fixed period t ,
 - (M, C, K, E_e, D_d)
 - K is the set of all permutations on $\{1, 2, \dots, t\}$
 - For each $e \in K$, the encryption function E_e is defined as follows:
 $E_e(m) = (m_{e(1)}, \dots, m_{e(t)})$
 $// m = (m_1, \dots, m_t) \in M$, the message space
 - The decryption key $d = e^{-1}$
 $//$ the inverse permutation of e
 - To decrypt the message $c = (c_1, \dots, c_t)$, compute
 $m = D_d(c) = (c_{d(1)}, \dots, c_{d(t)})$



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Transposition Ciphers



Plain: thisisatestofrailfence

Crypto: TIEFLCHSSTSORIFNEIATAE

Crypto: TAFEHSTORFNIETALCSSIE



Plain: thisatestoftheroutecipher

Reverse horizontal: TASINT TPOTSE TUOREH ENPICE VMXYR
Down vertical : TENER HSECE ITRIY SOOPX AFUHN TTTEV
Diagonal: T AT SPT IOUE HTOHV TSRPW EBIX HCY EZ R
Spiral counterclockwise: VETTT ASINT EHER ZYXN HUF OTS ECIPOR



Plain: thisisatestofcolumntransposition

Ciphertext: ITUNT~~X~~ IEORSZ HTCTON SSLAIY SOMSIW TAFNPO



- *Ciphertext only*:
 - adversary has only c ;
 - goal is to find plaintext, possibly key
- *Known plaintext*:
 - adversary has m, c ;
 - goal is to find the key
- *Chosen plaintext*:
 - adversary may get a specific plaintext enciphered;
 - goal is to find the key
- *Chosen ciphertext*:
 - adversary specifies a ciphertext, C , and gets the plaintext;
 - goal is to find the key
- Rubber-hose attack:
 - Adversary breaks knuckles, blackmails, threatens or tortures someone until they reveal the key;



- Cryptanalysis of classical ciphers relies on redundancy in the source language (plaintext)
 - E appears far more frequently than Z
 - Q is followed by U
 - TH is a common digram
 - Repeated letters in **pattern words** (e.g., three, soon, etc.)
- It's called statistical attacks, i.e., based on language statistics
- Brute force (exhaustive search):
 - Tries all possible keys on a piece of ciphertext
 - If the *number of keys* is small, then it's easy to break the encryption